

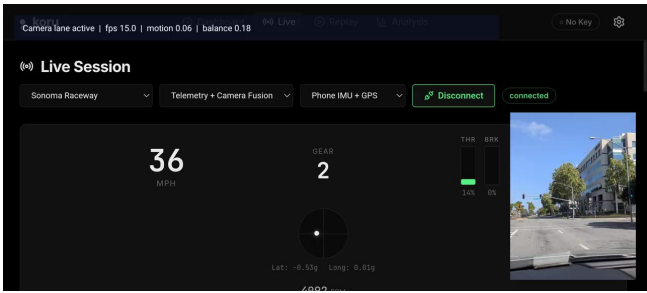
A Domain Expertise Layer on top of the Google stack

Today's best telemetry coaches (Garmin Catalyst) tell you what went wrong with numbers, after the fact. This system delivers context-aware coaching *as it happens*, adapted to driver skill, by routing every decision through a layered domain doctrine — Ross Bentley pedagogy, T-Rod's Sonoma session notes, and corner-by-corner Sonoma doctrine — before any LLM ever speaks. **Gemini becomes a tool inside a trust framework, not the source of trust.**

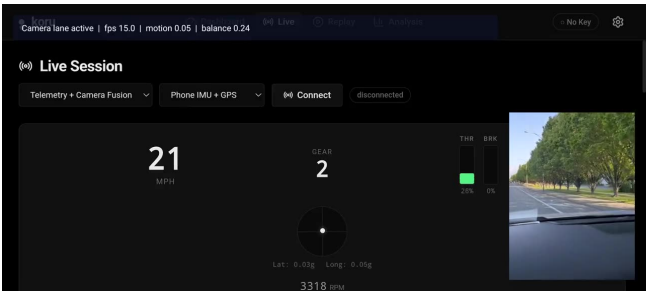
Owner Team 1	Branch consolidated	Tests 60 / 60 passing (13 files)	Stack Gemma 4 E2B-it (LiteRT-LM) · Gemini 3.1 Pro / 3 Flash preview Gemini 2.5 Flash Lite · Gemini 2.5 Pro TTS · Pixel 10 · CameraX 1.4.2 · MediaPipe Tasks GenAI 0.10.27
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Live demo — captured on Pixel 10

Two screen captures from the consolidated build. Telemetry is the master clock; the latest CameraX vision frame is fused onto each telemetry frame; the DEL shapes everything that comes out the speaker. Embedded MP4 versions ship in the design-assets folder alongside this PDF.



Demo 1 · Sonoma Raceway · Telemetry + Camera Fusion
54s · Phone IMU + GPS source · 36 mph, Gear 2, Lat -0.53g during a left-hander. Vision lane active (fps 15, motion 0.06).
media/demo-sonoma-live.mp4



Demo 2 · Device Camera + GPS Test → Telemetry + Camera Fusion
2m21s · Trackless device test transitioning to fused mode at 21 mph. Validates the camera pathway independently from telemetry. media/demo-camera-fusion.mp4

Three pillars of the framework

1. Domain Expertise Layer (load-bearing) Ross Bentley's 7 mental models + T-Rod's 5 Sonoma coaching patterns + corner-by-corner Sonoma doctrine. Not prompt text — it actively suppresses premature advice, biases hot-path wording, and rewrites feedforward per corner and per skill level. Doctrine, not decoration.	2. Split-Brain Coaching Engine Hot path (12-rule DecisionMatrix, <50ms, no I/O). Cold path (gemini-2.5-flash-lite, 2–5s). Edge reasoner cascade: AICore (Gemma 4 preview) → LiteRT-LM (Gemma 4 E2B-it) → Deterministic. The hot path NEVER waits for the cold path. P0 safety preempts everything.	3. Trust Gates Before Speech Timing state machine (OPEN → DELIVERING → COOLDOWN → BLACKOUT during apex). Driver model classifies skill from input smoothness. Bench-safe motion gating suppresses stationary false positives. Confidence thresholds gate every utterance.
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Verified tech stack — exact models, exact versions

Pulled directly from `pixel-android-app/app/build.gradle.kts`, `models/current-model.json`, `useGeminiCloud.ts`, `geminiService.ts`, and `coachingService.ts`.

Layer	Component	Exact identifier in code
On-device LLM · tier 1	Gemini Nano 4 / Gemma 4 preview via Android AICore Prompt API	<code>com.google.ai.edge.aicore.GenerativeModel</code> <code>runtime/reasoner/AiCoreReasoner.kt</code>
On-device LLM · tier 2 (active)	Gemma 4 E2B-it via LiteRT-LM (MediaPipe Tasks GenAI)	<code>litert-community/gemma-4-E2B-it-litert-lm</code> <code>file: gemma-4-E2B-it.litertlm</code> <code>sha256: ab7838cd...b27e42 (verified)</code> <code>device path: /data/local/tmp/koru/models/gemma-4-e2b-it/</code>
On-device LLM · tier 3 (floor)	Deterministic-only — phrase catalog + 12-rule DecisionMatrix, no LLM	<code>runtime/reasoner/DeterministicOnlyReasoner.kt</code> guarantees the system never goes silent on ML failure
Cloud cold path	gemini-2.5-flash-lite via REST v1beta	<code>https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash-lite:generateContent</code> <code>services/coachingService.ts:650</code>
Post-session (deep)	gemini-3.1-pro-preview with ThinkingLevel.HIGH	<code>hooks/useGeminiCloud.ts:14, 99–104</code> via <code>@google/genai 1.45.0</code>
Post-session (fast)	gemini-3-flash-preview with ThinkingLevel.LOW	<code>hooks/useGeminiCloud.ts:15, 99–104</code>
TTS	gemini-2.5-pro-preview-tts · <code>@google/genai v1alpha</code> streaming	<code>responseModalities: [Modality.AUDIO]</code> <code>hooks/useGeminiCloud.ts:124–127,</code> <code>useTTS.ts:83</code>
Android	Kotlin · MediaPipe Tasks GenAI 0.10.27 · CameraX 1.4.2	<code>minSdk 29 · target 35 · JVM 17</code> <code>com.google.mediatask:tasks-genai:0.10.27</code> <code>androidx.camera:camera-* 1.4.2 · webkit 1.12.1</code>
Web	React 19.2 · TypeScript 5.9 · Vite 8 · Tailwind 4 · Recharts · Papaparse	<code>@google/genai 1.45.0 · Vitest 4.1</code> <code>13 test files · 60 / 60 passing</code>

Three-tier on-device reasoner cascade. `EdgeRuntimeManager.warmupPreferredBackend()` probes each backend in order. The first one that responds with a non-UNAVAILABLE / non-ERROR state becomes the active reasoner; the choice is persisted to `SharedPreferences`. Confidence per tier: AICore **0.78**, LiteRT-LM **0.72**, Deterministic **baseline**. The deterministic floor is what makes this trustable — the system never goes silent because of an ML failure.

Tier 1 · AICore Gemini Nano 4 / Gemma 4 preview Prompt API. Probed first via reflection ON <code>com.google.ai.edge.aicore.GenerativeModel</code> . Highest confidence (0.78). Supports HOT, FEEDFORWARD, EDGE paths.	Tier 2 · LiteRT-LM Gemma 4 E2B-it (.litertlm, sha256-verified). Loaded via MediaPipe <code>tasks.genai.llminference.LlmInference</code> . Pushed to device by <code>npm run pixel:e2e:prepare</code> .	Tier 3 · Deterministic Phrase catalog + 12-rule decision matrix. No LLM. Always available. This is the trustable floor — the system never goes silent due to ML failure.
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Domain Expertise Layer — what's actually in it

Implemented in `src/data/trackExpertise.ts`, `src/data/trodCoachingData.ts`, and `src/utils/coachingKnowledge.ts`. Wired into the runtime selector, not just the prompt.

Pedagogy / Doctrine source	What it contributes at runtime
Ross Bentley pedagogy (7 mental models) Friction Circle · Weight Transfer · Trail Braking · Vision Drives the Car · Maintenance Throttle · Slow In Fast Out · One Thing at a Time	Skill-adapted humanization, beginner trigger phrases ('Hard initial!', 'Squeeze, don't stab', 'Eyes up!'), session-progression suppression of advanced techniques in the first 60s.
T-Rod Sonoma coaching (5 patterns) Throttle Commitment · Brake Trace Quality · Delay Early Throttle · Distance Is King · Session Learning Sequence	Hot-path rules (EARLY_THROTTLE, LIFT_MID_CORNER, SPIKE_BRAKE) and corner-specific feedforward wording on Sonoma sweepers.
Sonoma track doctrine Corner dossiers for T2, T3, T3A, T6, T7, T9–10, T11, T12; three timing sectors with distinct guidance.	Feedforward triggers tied to corner geometry; sector-specific advice replaces generic LLM output. Sonoma is the primary deployed track in both web registry and native Android catalog.
Driver model + session goals 10s rolling window, 5s hysteresis, max 3 pre-race goals.	Beginner: feel-based feedback. Intermediate: persona phrasing. Advanced: data-driven (G-Lat, exit speed delta). Goals bias hot-path rule activation.

“The right message at the wrong time is the wrong message. Feedback 800ms late is worse than silence.” — Brian Luc

Split-Brain Coaching Engine

Path	Latency	Backed by	Tier	Role
Hot	< 50 ms	12-rule DecisionMatrix + DriverModel	P0/P1/P3	Per-frame act-now coaching. Rules: OVERSTEER_RECOVERY, THRESHOLD, TRAIL_BRAKE, COMMIT, THROTTLE, PUSH, COAST, HESITATION, FULL_THROTTLE, EARLY_THROTTLE, LIFT_MID_CORNER, SPIKE_BRAKE.
Cold	2–5 s	gemini-2.5-flash-lite via REST v1beta, skill-adapted prompt	P2	Multi-frame physics analysis. 10 / 20 / 15-word ceiling per skill. Enriches future hot-path; never blocks.
Feedforward	trigger	Geofence + Sonoma T-Rod corner dossiers	P1	Anticipatory advice 150m before corner. Per-corner wording from T-Rod notes.
Edge	on-device	AlCore (Gemma 4 prev) → LiteRT-LM (Gemma 4 E2B-it) → Deterministic	P1/P2	Three-tier cascade. Auto-warms; persisted to SharedPreferences.

Post	5–30 s	gemini-3.1-pro-preview (HIGH) · gemini-3-flash-preview (LOW)	review	Sector-by-sector lap analysis with explicit ThinkingLevel. Runs on saved artifact, not live.
TTS	50–200 ms	gemini-2.5-pro-preview-tts · @google/genai v1alpha	output	Persona voice. generateContentStream + Modality.AUDIO. Web Speech API fallback.

Trust gates that fire before any utterance

Gate	Status	What it does
Timing state machine	Done	OPEN→DELIVERING→COOLDOWN→BLACKOUT. Beginners: 3s cooldown, mid-corner blackout. Advanced: 1s, no blackout. P0 bypasses all.
Priority queue + preempt	Done	P0 preempts all. Max 5 items, 3s stale expiry. Safety alerts never sit behind technique tips.
Driver model + hysteresis	Done	10s rolling window (rate-agnostic 8–25Hz). 5s hysteresis. Smoothness < 0.4 OR coasting > 30% ⇒ BEGINNER.
Bench-safe motion gating	Done	Phone-only telemetry coaching suppressed until real movement detected. Eliminates stationary false positives.
Session phase filter	Done	Phase 1 (0–60s) suppresses TRAIL_BRAKE, COMMIT, ROTATE, EARLY_THROTTLE. Advanced drivers skip to Phase 3.
Confidence + speed gates	Done	Realtime speech hardening. Higher-priority cues interrupt lower-priority speech. Faster timing for advanced.
Pre-rendered safety MP3s	Planned	AudioContext pre-caching wired; need per-persona BRAKE/OVERSTEER clips to bypass TTS latency.
Cold path offline fallback	Future	Pre-computed coaching lookup table per known track. Pixel 10 + Gemma 4 evaluated as Nano upgrade.

How the Google stack plugs in

Layer	Google component	Exact identifier in code	Status
Compute / OS	Pixel 10 · Android (minSdk 29, target 35)	KoruTelemetryService foreground service · JVM 17	Done
Vision	CameraX 1.4.2	camera/CameraLaneAnalyzer.kt · VisionFeatureStore.kt	Done
On-device runtime	MediaPipe Tasks GenAI 0.10.27	com.google.mediapipe:tasks-genai:0.10.27 · LlmInference	Done
On-device LLM (preferred)	Gemini Nano 4 / Gemma 4 preview · AICore	com.google.ai.edge.aicore.GenerativeModel	Scaffolded
On-device LLM (active)	Gemma 4 E2B-it via LiteRT-LM	litert-community/gemma-4-E2B-it-litert-lm · sha256-verified	Done
Cold path	gemini-2.5-flash-lite	v1beta REST · coachingService.ts:650 · geminiService.ts:30	Done
Post-session deep	gemini-3.1-pro-preview, ThinkingLevel.HIGH	hooks/useGeminiCloud.ts:14 · @google/genai 1.45.0	Done
Post-session fast	gemini-3-flash-preview, ThinkingLevel.LOW	hooks/useGeminiCloud.ts:15	Done
TTS	gemini-2.5-pro-preview-tts + Web Speech API	generateContentStream · Modality.AUDIO · v1alpha	Done
Storage	BigQuery / GCS (AGY pipeline)	Future post-session schema for cross-session learning	Planned

Latency budget — telemetry event to spoken coaching

Stage	Latency	Notes
RaceBox BLE → Pixel 10	7–15 ms	BLE 5.2 high-priority connection
OBD K-Line round-trip	80–150 ms	Vehicle-side, cannot reduce
Android BLE → fusion	5–10 ms	Foreground service
AI inference (hot path)	< 50 ms	Heuristic rules, no I/O
TTS audio output	50–200 ms	Web Speech / Gemini TTS
TOTAL BUDGET	~200–425 ms	Under Brian Luc's 800ms 'worse than silence' threshold

Cross-pollination opportunities for the cohort

These are the patterns we believe should be lifted to cohort-level standards. The DEL shape transfers to any domain (medical, ops, fraud); the racing specifics don't.

Opportunity	Status	Why it matters across teams
VehicleDataStream contract	Proposed	Abstract OBD/CAN/BLE behind one interface. Coaching engine never changes when upgrading from OBD to direct CAN. Ready for Team 2 (CAN-to-USB) and Team 3 (MoTec).
Shared post-session schema	Open	Per-corner metrics, mistake zones, coaching decisions. Codify the saved-session artifact format so AGY pipeline ingestion is identical across the three teams.
Priority tier convention	Reusable	P0 safety / P1 tactical / P2 strategic / P3 encouragement. Trivially portable. Recommended cohort default for any real-time advisory work.
Time-based windows over frame counts	Reusable	Sensor rates differ (8 Hz OBD vs 25 Hz RaceBox). Always store {time, value} pairs and trim by elapsed time. One decision eliminated an entire bug class.
Domain Expertise Layer pattern	Reusable	Doctrine that filters and shapes LLM output is the trust mechanism. Required between any raw model and any user-facing output.
Mocked telemetry endpoint	In progress	Throttled synthetic stream so all teams validate ingestion pre-field-test. Owned by Team 1 — adopting it unblocks pipeline development for everyone.

Team 1 — Beginner Pod

The six members behind this submission.

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Mentors & doctrine sources (not Team 1 members): Ross Bentley (Speed Secrets author, Garmin Catalyst consultant — pedagogy); T-Rod / Tony Rodriguez (real Sonoma coaching session — track-specific patterns); Brian Luc (Edge mentor — *"Feedback 800ms late is worse than silence."*).

Repository & supporting documents

Asset	Link / Path
GitHub — Team 1 primary repository ('consolidated' branch)	https://github.com/rabimba/trustable-ai-codelab
GitHub — upstream cohort reference (ykro)	https://github.com/ykro/trustable-ai-codelab
Interactive design asset (HTML, embedded video demos)	design-assets/trustable-ai-framework.html
Demo video 1 — Sonoma Live · Telemetry + Camera Fusion	design-assets/media/demo-sonoma-live.mp4
Demo video 2 — Device Camera + GPS Test → Fusion	design-assets/media/demo-camera-fusion.mp4
End-to-End Architecture	docs/end-to-end-architecture.md
Data Reasoning module reference	docs/data-reasoning.md
Real-Time Reasoning Learnings (portable patterns)	docs/learnings-real-time-data-reasoning.md
Consolidation Report (what was merged from where, attribution)	docs/consolidation-report.md
Implementation history	docs/implementation-history.md
Current implementation audit	docs/current-implementation-audit.md
User stories	docs/user-stories.md
Pre-race chat contract	docs/pre-race-chat-contract.md

Attribution. Submitted by **Team 1** (Beginner Pod). Member list on page 9. Domain coaching content is sourced from Ross Bentley, T-Rod (Tony Rodriguez), and Brian Luc and is integrated as doctrine, not modified.

Built with the Google stack · Pixel 10 · CameraX · Gemini · LiteRT-LM
